COLLEGE OF INFORMATION TECHNOLOGY AND COMPUTING
Data Science Department

SYLLABUS
Course Title: Computer Programming
Course Code: DS112
Credits: **3 units (2 hours Lecture, 3 hours Laboratory)**

USTP Vision
A nationally - recognized Science and Technology University providing the vital link between education and the economy.

USTP Mission
- Bring the world of work (industry) into the actual higher education and training of students;
- Offer entrepreneurs the opportunity to maximize their business potentials through a gamut of services from product conceptualization to commercialization;
- Contribute significantly to the national development goals of food security and energy sufficiency through Technology solutions

Year & Semester: **First Year, 1st Semester AY 2026-2027**
Class Schedule: (LAB & LEC)
DS1A: T 6:00 PM - 9:00 PM | W 3:00 PM - 5:00 PM
DS1B: S 1:00 PM - 4:00 PM | T 5:00 PM - 7:00 PM
DS1C: M 10:00 AM - 1:00 PM | W 11:00 AM - 1:00 PM
DS1D: Th 1:00 PM - 4:00 PM | W 9:00 AM - 11:00 AM
Building/Room #: See approved PRISMS classroom assignment

Prerequisite(s): **None**
Co-requisite(s): **None**

Instructors:
DS1A - MATTHEW R. MAULION | matthewr.maulion@ustp.edu.ph
DS1B - TIMOTHY JONAH E. BORROMEO | timothyjonah.borromeo@ustp.edu.ph
DS1C - JHUN BRIAN M. ANDAM | jhunbrian.andam@ustp.edu.ph
DS1D - IAN JURE E. MACALISANG | ian.macalisang@ustp.edu.ph
Mobile/Contact No.: local 1228

Consultation Schedule: **WED | 1 PM to 5 PM**
Bldg./Rm. No.: **DS Office | 9-204**
Office Phone No./Local: **local 1228**

I. Course Description:
This course introduces computer programming using the Python programming language where students will gain the requisite computing skills of a data science practitioner. These computing skills allow data scientists to handle different kinds of data stored in different formats and media using available or custom code. Emphasis is placed on procedural programming, algorithm design, and language constructs common to most high-level languages. Students will learn how to write readable, understandable and PEP8-compliant code in Python using the PyData stack. Acquired skills in this course will lay the foundations for advanced succeeding data science courses in data mining, machine learning, and computational statistics.

Program Educational Objectives:

Three (3) to Five (5) years after graduation, BSDS graduates are:

PE01. Proficient in the Data Science field and able to engage constantly in big data analysis and professional advancement by pursuing a higher academic level and/or practicing quality improvement in their career or entrepreneurial endeavor;

PE02. Highly competent in generating new ideas and innovations in Data Science emphasizing on capturing, storing, retrieving and visualizing massive data; and

PE03. leading data scientists who can effectively work on data sets to extract knowledge and identify patterns in order to predict trends; and contribute significantly to human development, socio-economic transformation, and national initiatives.

II. Course Outcomes(CO)	**Program Outcomes (PO)**													
	a	**b**	**c**	**d**	**e**	**f**	**g**	**h**	**i**	**j**	**k**	**l**	**m**	**n**
CO1: Design, implement, test, and debug readable Python programs using algorithms, variables, expressions, control structures, and core language constructs.	D	D	E	E	E		I	I		I		I	I	
CO2: Develop modular and well-documented programs using functions, modules, introductory classes, collections, and file and exception handling.	D	E	E	E	E	I	E	E	I	E		I	E	
CO3: Apply NumPy and pandas to acquire, transform, aggregate, and analyze structured data and present a tested, reproducible PyData programming solution.	D	E	E	E	E	E	E	D	D	D	E	I	E	I

III. Course Outline:

Allotted Time	**Course Outcomes (CO)**	**Intended Learning Outcomes (ILO)**	**Topic(s)**	**Suggested Readings**	**Teaching-Learning Activities**	**Assessment Tasks/Tools**	**Grading Criteria**	**Remarks**
2HRS W1 August 10-15, 2026	CO1	Explain course expectations, fundamental programming concepts, and the role of Python in data science.	**COURSE ORIENTATION AND PROGRAMMING FOUNDATIONS** 1. Course Syllabus, Class policies, grading system Presentation 2. University Vision and Mission	*USTP Student Handbook* *Course Syllabus*				

Program Outcomes:
Upon completion of the program, BSDS graduates are able to:

a. Apply knowledge of computing science, and mathematics, and business management in solving complex data-driven problems;

b. Use current standards and best practices within data-science and specific areas of mathematics (e.g. statistical analysis, optimization, machine learning, network analysis, experiment design, and algorithms, among others.) in solving complex data-driven problems and requirements;

c. Analyze complex data-driven problems by applying analytical and quantitative reasoning; and identify the statistical and computing requirements appropriate to its solution;

d. Extract, process and analyze data sets, including Big

			3. CITC and Department orientation 4.					
13 HRs W1-W3 August 10- 28, 2026		Determine the differences of programming Languages Create algorithms and flowcharts for programming problems Explore and discover Jupyter IDE functionalities Translate algorithmic logic in Jupyter Notebooks	**INTRODUCTION TO COMPUTER PROGRAMMING AND THE JUPYTER NOTEBOOK** - PROGRAMMING LANGUAGES - ALGORITHMS AND FLOWCHARTS - JUPYTER NOTEBOOK AS IDE	*DS 112 Computer Programming (Instructional Material) – Maulion, Matthew R.* *Introduction to Programming Using Python* *www.tutorialspoint.com* *https://jupyter-notebook.readthedocs.io/en/stable/*	Lecture/ Discussion Group Break-out sessions	Release of Seatworks · Algorithm making Jupyter Notebook Exercises Oral recitation Release of Assignment 1 (Reflection Paper)	See Grading System	Lecture/ Laboratory
	CO1, CO2, CO3	Implement Program logic in Python	**PYTHON PROGRAMMING** 1. VARIABLES, EXPRESSIONS, & STATEMENTS 2. CONTROL STRUCTURES (ITERATION AND CONDITIONALS) 3. FUNCTIONS	*DS 112 Computer Programming (Instructional Material) – Maulion, Matthew R.*	Lecture/ Discussion Group Break-out sessions	Release of Seatworks - Coding problem exercises	See Grading System	Lecture/Laboratory

Datasets too large for traditional data processing techniques; e. Design, implement and evaluate different predictive models, techniques, processes, components, or programs to meet desired needs and requirements under various constraints; f. Integrate effectively the big data analytics solutions into government sectors such as transportation, public health and safety, environmental issues as well as issues in the corporate and non-profit organizations; g. Create, select, adapt and apply appropriate techniques, resources, mathematical or statistical models, machine/deep learning and other modern tools to complex data science activities; h. Utilize data science and its technologies as strategies in marketing/branding of potential entrepreneurial ventures;	25 HRs W4-W8 August 31- October 2, 2026		Define and call User-defined functions appropriate for a given programming problem Create data structures, utilize and differentiate its advantages and disadvantage against each other Define Objects and Classes in Python	4. Lists, Tuples, Strings 5. Dictionaries and Collections 6. Files (Exceptions) 7. Introduction to Objects and Classes	*Introduction to Programming Using Python* www.tutorialspoint.com		Oral recitation Release of Assignment 2, 3, and 4		
	5 HRs W9 (October 5-10, 2026)	**MIDTERM EXAMINATION**							
	15 HRs W10-W12 October 12- 30 , 2026	**CO1, CO2, CO3**	Familiarize and implement built-in functions from the	**THE NUMPY (ARRAY)** 1. Introduction and Drawbacks 2. Data types	*DS 112 Computer Programming (Instructional Material) – Maulion, Matthew R.*	Lecture/ Discussion Group Break-out sessions	Release of Seatworks · Coding problem exercises	See Grading System	Lecture/Laboratory

i. Function effectively as individual or work collaboratively and respectfully as a member or leader in diverse development teams and in multi-cultural and multi-disciplinary settings; j. Communicate effectively in both oral and in written form by being able to deliver and comprehend instructions clearly; and present persuasively to diverse audience the data science-related ideas and perspectives; k. Assess the model used to solve data science tasks and identify its local and global impact on individuals, organizations, and society; l. Recognize the need to engage in independent learning and be at pace with the latest development in a specialized Data Science field, with emphasis on Massive Data Analytics and Business			Numpy package Distinguish the upside and draw backs among 1D, 2D & Multidimensional Arrays Apply 1D, 2D & Multidimensional Arrays operations in simple programming problems	3. 1D Array Operations, Applications 4. Two-Dimensional Array and its Operations a. Insertion, deletion, searching, and sorting b. Concatenation, splitting, and broadcasting c. Universal functions d. Axis operations and matrix arithmetic e. Aggregation, reshaping, and transposition	*Numpy Documentation* *[https://numpy.org]*		Oral recitation Release of Assignment 6		
	20 HRs W13-W16 November 2- 27, 2026	**CO1, CO2, CO3**	Familiarize and implement basic built-in functions from the Pandas package Perform computational and	**PANDAS** 1. Introduction to Data Structures, Essential Basic Functionality 2. Series, DataFrames, indexes, and data types 3. Reading/writing CSV files, Working with Text Data	*DS 112 Computer Programming (Instructional Material) – Maulion, Matthew R.* *Pandas documentation*	Lecture/ Discussion Group Break-out sessions	Release of Seatworks · Coding problem exercises Oral recitation		

Intelligence for continual development as a computing professional;

m. Participate in generation of new knowledge, or in research and development projects with the end view of contributing to local and national economy; and

n. Preserve and promote "Filipino historical and cultural heritage".

		visualization tasks using Pandas Demonstrate aggregation and merging tools to toy and real-life datasets Implement pivot functionalities	4. Indexing, selection, and summary operations 5. Computational tools and Visualization 6. Group By, Aggregation, Transformation 7. Merge, join, concatenate, and other operations 8. Reshaping and Pivot Tables	*[http://pandas.pydata.org/pandas-docs/stable/]*		Release of Assignment 6		
5 HRs W17 (November 30 - December 4, 2026)	**FINAL PROJECT ORAL PRESENTATION**							
5 HRs W18 (December 7-19, 2026)	**FINAL EXAMINATION WEEK**							

IV. Course Requirements:

1. Class attendance and participation:

Attendance and Participation in Co-Curricular Activities

- Students are expected to attend and actively participate in relevant webinars, training, and competitions as part of their academic engagement. Participation in such activities will be awarded corresponding points, which shall be determined and arranged by the faculty and reflected in the overall grading system.

Refer to the current USTP Student Handbook for attendance and participation policies.

2. Course readings/materials:

(a)Books, E-books

1. Maulion M. (2022) DS 112 Computer Programming (Instructional Material)
2. Allen B. Downey, Think Python.
3. Jake VanderPlas, Python Data Science Handbook.
4. Y. Daniel Liang, Introduction to Programming Using Python.
5. Wes McKinney, Python for Data Analysis.
6. Current official documentation for Python, Jupyter Notebook, NumPy, and pandas.
7. Michael Churchmurti. (2019). Computer programming for beginners: a step-by-step guide to learn coding. [Paperback].
8. John Paul Mueller and Luca Massaron, (2017). Algorithms [Book / Paperback]. Hoboken, New Jersey:John Wiley & Sons, Inc.
9. Dan Zhang. (2017), R for programmers: advanced techniques | Book | Paperback] Boca Raton : CRC Press; Taylor & Francis Group,
10. Herbert Welpo , (2017), Introductory statistics using SPSS® Los Angeles : Sage Publications Inc
11. Song S.Q.,(2017), Environmental and ecological statistics with R.[Book ; Hardcover] Boca Raton : CRC Press,
12. John Paul Mueller and Luca Massaron. (2020) Data Science programming all-in-one. Hoboken, New Jersey: John Wiley & Sons, Inc.
13. Ron S. Kenett, Thomas C. Redman (2019). The Real work of Dat Science: Turning data into information, better decisions, and stronger organizations. Hoboken, NJ: John Wiley & Sons, Inc.
14. Norman K. Denzin. (2018). The Sage Handbook of Qualitative research / [Book/ Hardcover]. Los Angeles: Sage.
15. David R. Anderson, (2018). Essentials of statistics for businesss & economics. Australia: [Book/Paperback]. Amsterdam : Morgan Kaufmann.
16. Geoffrey Cumming, Robert Calin-Jageman. (2017). Introduction to the new statistics: estimation, open science, and beyond / [Book/Picture Book]. New York : Routledge, Taylor & Francis Group
17. Timothy C. Urdan (2017). Statistics in plain English / [Book / Paperback]. New York, NY: Routledge, Taylor & Francis Group.
18. Jose Manuel Magallanes Reyes (2017). Introduction to Data Science for Social and Policy Research: Collecting and organization data with R and python. Cambridge, United Kingdom; New York, NY:Cambridge University Press.
19. Approved institutional instructional materials and laboratory guides.

(b) Supplies needed (desktop/laptop computer, stable Internet connection, and Python/Jupyter environment)

3. Assignments, Assessment, and Evaluation

(a) Policy concerning homework

- Homework and laboratory outputs shall be submitted on the scheduled dates. Late submissions follow the Student Handbook and documented course arrangements.
- Required laboratory exercises, quizzes, portfolios, and project milestones shall be completed throughout the semester.

(b) Policy concerning make-up exams

- Make-up examinations are granted only for valid, documented reasons and are scheduled in accordance with University policy.

(c) Policy concerning special quizzes

- Special quizzes are granted only to students who present valid reasons and appropriate supporting documents for their absence.
- The special quiz shall be scheduled according to the instructor's documented arrangement and University policy.

(d) Preliminary information on term papers or projects, with due dates

- Late project submissions shall be handled in accordance with documented course policies and approved arrangements.

(e) List of requirements that will impact the final grade

- Programming workshops, quizzes, midterm and final examinations, NumPy and pandas portfolios, and the integrated PyData project.

(g) GRADING POLICY

(Passing Percentage is 70%)

Ex. In a 10-item quiz, obtaining 7 points would be equivalent to a passing score.

(h) Guidelines on the Ethical Use of AI Tools

- Students may use AI-powered tools (e.g., ChatGPT, Grammarly, or statistical software with AI features) only as assistive technologies, such as for brainstorming ideas, improving writing clarity, or supporting data analysis.
- Any use of AI must be properly disclosed, either in the references or through a brief AI Acknowledgement (e.g., "ChatGPT was used to rephrase draft text and suggest structure; the final work is the student's own.").
- Students are not permitted to use AI to generate entire assignments, research papers, or exam answers, as this constitutes academic dishonesty.
- Students remain fully responsible for verifying the accuracy, originality, and ethical integrity of any AI-assisted output.
- Any misuse of AI—including plagiarism, data fabrication, or over-reliance without critical evaluation—will be treated as a violation of academic integrity.

Grading System

Lecture Grade (67%)	
Performance Item/Criteria	***%***
Class Performance Item	10%
Quizzes (All quizzes, prelim and pre-final exams)	40%
Major Exams (i.e, Midterm and Final Exams)	30%
Performance Innovative Task / Project	20%
TOTAL	**100%**
Laboratory Grade (33%)	
Performance Item/Criteria	***%***

Laboratory Exercises/Reports	30%
Laboratory Major Exam	40%
Hands on Exercises	30%
TOTAL	**100%**
Term/Periodic Grade = 67% Lecture Grade + 33% Laboratory Grade	
Options: **FINAL GRADE (FG) = 1/3 Midterm Grade (MTG)+ 2/3 Final Term Grade (FTG)** **FINAL GRADE (FG) = 1/2 Midterm Grade (MTG)+ 1/2 Final Term Grade (FTG)**	

- The use of **USTeP (University of Science and Technology E-learning Portal)** as the official Learning Management System To distribute course materials, to communicate and collaborate online, to post grades, to submit assignments, and to give you online quizzes and surveys.
- Rubrics Used for Assessment
- **Journal:**

Criteria	Unsatisfactory-Beginning	Developing	Accomplished	Exemplary	Total
Content Reflection	**0-7 points**	**8-11 points**	**12-16 points**	**17-20 points**	/20
	Reflection lacks critical thinking. Superficial connections are made with key course concepts and course materials, activities, and/or assignments	Reflection demonstrates limited critical thinking in applying, analyzing, and/or evaluating key course concepts and theories from readings, lectures, media, discussions, activities, and/or assignments Minimal connections made through explanations, inferences, and/or examples.	Reflection demonstrates some degree of critical thinking in applying, analyzing, and/or evaluating key course concepts and theories from readings, lectures, media, discussions activities, and/or assignments. Connections made through explanations, inferences, and/or examples.	Reflection demonstrates a high degree of critical thinking in applying, analyzing, and evaluating key course concepts and theories from readings, lectures, media, discussions activities, and/or assignments. Insightful and relevant connections made through contextual explanations, inferences, and examples.	
Personal Growth	**0-4 points**	**5-7 points**	**8-11 points**	**12-15 points**	/15
	Conveys inadequate evidence of reflection on own work in response to the self-assessment questions posed. Personal growth and awareness are not evident and/or demonstrates a neutral experience with negligible personal impact. Lacks enough inferences, examples, personal insights and challenges, and/or future implications are overlooked.	Conveys limited evidence of reflection on own work in response to the self-assessment questions posed. Demonstrates less than adequate personal growth and awareness through few or simplistic inferences made, examples, insights, and/or challenges that are not well developed. Minimal thought of the future implications of current experience.	Conveys evidence of reflection on own work with a personal response to the self-assessment questions posed. Demonstrates satisfactory personal growth and awareness through some inferences made, examples, insights, and challenges. Some thought of the future implications of current experience.	Conveys strong evidence of reflection on own work with a personal response to the self-assessment questions posed. Demonstrates significant personal growth and awareness of deeper meaning through inferences made, examples, well developed insights, and substantial depth in perceptions and challenges. Synthesizes current experience into future implications.	
Writing Quality	**0-4 points**	**5-7 points**	**8-11 points**	**12-15 points**	/15

	Poor writing style lacking in standard English, clarity, language used, and/or frequent errors in grammar, punctuation, usage, and spelling. Needs work.	Average and/or casual writing style that is sometimes unclear and/or with some errors in grammar, punctuation, usage, and spelling.	Above average writing style and logically organized using standard English with minor errors in grammar, punctuation, usage, and spelling.	Well written and clearly organized using standard English, characterized by elements of a strong writing style and basically free from grammar, punctuation, usage, and spelling errors.	
Timeliness	**Deduct 11 points-overall failing**	**Deduct 6-10 points**	**Deduct 1-5 points**	**0 points deducted**	**/-**
	Journal reflection is submitted 2-3 days (49-72 hours) after the deadline.	Journal reflection is submitted 1-2 days (25-48 hours) after the deadline.	Journal reflection is submitted within 1 day (24 hours) after the deadline.	Journal reflection is submitted on or before deadline.	
				TOTAL POINTS (sum of 4 Criteria)	**/50**

- **Final Project:**

CATEGORY	5	4	2	1
Organization	Information is very organized in a well-constructed paragraph or paragraphs.	Information is organized in a well-constructed paragraph or paragraphs.	Information is organized, but paragraph(s) are not well-constructed.	The information appears to be disorganized.
Amount of Information	All topics are addressed and all questions answered with at least 2 sentences about each.	All topics are addressed and most questions answered with at least 2 sentences about each.	All topics are addressed, and most questions answered with 1 sentence about each.	One or more topics were not addressed.
Quality of Information	Information clearly relates to the topic. It includes details and/or examples.	Information relates to the topic. It provides few supporting details and/or examples.	Information partially relates to the topic. No details and/or examples are given.	Information has little or nothing to do with the topic.
Application of Learning	Information presented shows a clear understanding and application of the Professional Development Topic.	Information presented shows understanding and application of the Professional Development Topic.	Information presented shows a partial understanding and application of the Professional Development Topic.	Information presented shows little or no understanding and application of the Professional Development Topic.
Mechanics	No grammatical, spelling or punctuation errors.	Almost no grammatical, spelling or punctuation errors	A few grammatical spelling, or punctuation errors.	Many grammatical, spelling, or punctuation errors

- **Integrated PyData Project and Presentation**

PRESENCE -body language & eye contact -contact with the public -poise -physical organization	5	4	3	2	1	0
LANGUAGE SKILLS -correct usage -appropriate vocabulary and grammar -understandable (rhythm, intonation, accent) -spoken loud enough to hear easily	5	4	3	2	1	0
ORGANIZATION -clear objectives -logical structure -signposting	5	4	3	2	1	0
MASTERY OF THE SUBJECT -pertinence -depth of commentary -spoken, not read -able to answer questions	5	4	3	2	1	0
VISUAL AIDS -transparencies, slides -handouts -audio, video, etc.	5	4	3	2	1	0
OVERALL IMPRESSION -very interesting / very boring -pleasant / unpleasant to listen to -very good / poor communication	5	4	3	2	1	0

TOTAL SCORE: _____ / 30

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